```
# Install required libraries
!pip install transformers
!pip install torch

# Import libraries
from transformers import pipeline

print("Libraries installed successfully!")
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transfor
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Libraries installed successfully!
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# English → French
model_name_en_fr = "Helsinki-NLP/opus-mt-en-fr"
tokenizer_en_fr = AutoTokenizer.from_pretrained(model_name_en_fr)
model_en_fr = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_fr)

# English → German
model_name_en_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_en_de = AutoTokenizer.from_pretrained(model_name_en_de)
model_en_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_de)

# English → Hindi
model_name_en_hi = "Helsinki-NLP/opus-mt-en-hi"
tokenizer_en_hi = AutoTokenizer.from_pretrained(model_name_en_hi)
model_en_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_hi)

print("All models and tokenizers loaded successfully!")
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pi
  warnings.warn("Recommended: pip install sacremoses.")
Loading weights: 100%|| 258/258 [00:00<00:00, 571.72it/s, Materializing param=model.shared.weight]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weigh
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weigh
config.json: 1.33k/?|| [00:00<00:00, 54.8kB/s]
tokenizer_config.json: 100%|| 42.0/42.0 [00:00<00:00, 2.02kB/s]
source.spm: 100%|| 768k/768k [00:00<00:00, 11.5MB/s]
target.spm: 100%|| 797k/797k [00:00<00:00, 39.3MB/s]
vocab.json: 1.27M/?|| [00:00<00:00, 27.3MB/s]
pytorch_model.bin: 100%|| 298M/298M [00:04<00:00, 107MB/s]
Loading weights: 100%|| 258/258 [00:00<00:00, 583.93it/s, Materializing param=model.shared.weight]
model.safetensors: 100%|| 298M/298M [00:05<00:00, 72.9MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weigh
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weigh
generation_config.json: 100%|| 293/293 [00:00<00:00, 2.82kB/s]
config.json: 1.39k/?|| [00:00<00:00, 30.4kB/s]
tokenizer_config.json: 100%|| 44.0/44.0 [00:00<00:00, 755B/s]
source.spm: 100%|| 812k/812k [00:00<00:00, 36.3MB/s]
target.spm: 100%|| 1.07M/1.07M [00:00<00:00, 36.1MB/s]
vocab.json: 2.10M/?|| [00:00<00:00, 29.1MB/s]
pytorch_model.bin: 100%|| 306M/306M [00:03<00:00, 157MB/s]
Loading weights: 100%|| 258/258 [00:00<00:00, 568.46it/s, Materializing param=model.shared.weight]
model.safetensors: 100%|| 306M/306M [00:03<00:00, 113MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weigh
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weigh
generation_config.json: 100%|| 293/293 [00:00<00:00, 26.6kB/s]
All models and tokenizers loaded successfully!
```

```
sentences = [
    "I am learning artificial intelligence.",
    "The weather is very pleasant today.",
    "She completed her assignment on time.",
    "Can you help me with this problem?",
    "Machine learning is transforming industries.",
    "I went to the market and bought vegetables.",
    "Although it was raining, we continued our journey.",
    "What is your favorite programming language?",
    "The meeting will start at 10 AM tomorrow.",
    "Data science involves statistics and programming.",
    "He plays cricket and also studies well.",
    "If you work hard, you will succeed.",
    "This smartphone has advanced features.",
    "Let us discuss the project later.",
    "Do you understand this concept?"
]

print("Sentences loaded successfully!")
```

```
Sentences loaded successfully!
```

```
def translate(text, tokenizer, model):
    inputs = tokenizer(text, return_tensors="pt", padding=True)
    outputs = model.generate(**inputs)
    return tokenizer.decode(outputs[0], skip_special_tokens=True)
```

```
print("\n--- English to French ---\n")

for sent in sentences:
```

```
    translated = translate(sent, tokenizer_en_fr, model_en_fr)
    print("EN:", sent)
    print("FR:", translated)
    print()
```

```
--- English to French ---

EN: I am learning artificial intelligence.
FR: J'apprends l'intelligence artificielle.

EN: The weather is very pleasant today.
FR: Le temps est très agréable aujourd'hui.

EN: She completed her assignment on time.
FR: Elle a terminé sa mission à temps.

EN: Can you help me with this problem?
FR: Pouvez-vous m'aider avec ce problème ?

EN: Machine learning is transforming industries.
FR: L'apprentissage automatique transforme les industries.

EN: I went to the market and bought vegetables.
FR: Je suis allé au marché et j'ai acheté des légumes.

EN: Although it was raining, we continued our journey.
FR: Même s'il pleuvait, nous avons poursuivi notre voyage.

EN: What is your favorite programming language?
FR: Quel est votre langage de programmation préféré?

EN: The meeting will start at 10 AM tomorrow.
FR: La réunion commencera demain à 10 heures.

EN: Data science involves statistics and programming.
FR: La science des données implique des statistiques et des programmes.

EN: He plays cricket and also studies well.
FR: Il joue au cricket et étudie aussi bien.

EN: If you work hard, you will succeed.
FR: Si vous travaillez dur, vous réussirez.

EN: This smartphone has advanced features.
FR: Ce smartphone a des fonctionnalités avancées.

EN: Let us discuss the project later.
FR: Discutons du projet plus tard.

EN: Do you understand this concept?
FR: Comprenez-vous ce concept ?
```

```
print("\n--- English to German ---\n")

for sent in sentences:
    translated = translate(sent, tokenizer_en_de, model_en_de)
    print("EN:", sent)
    print("DE:", translated)
    print()
```

```
--- English to German ---

EN: I am learning artificial intelligence.
DE: Ich lerne künstliche Intelligenz.

EN: The weather is very pleasant today.
DE: Das Wetter ist heute sehr angenehm.

EN: She completed her assignment on time.
DE: Sie hat ihren Auftrag rechtzeitig erfüllt.

EN: Can you help me with this problem?
DE: Können Sie mir bei diesem Problem helfen?

EN: Machine learning is transforming industries.
DE: Maschinelles Lernen verändert die Industrie.
```

```
EN: I went to the market and bought vegetables.
DE: Ich ging auf den Markt und kaufte Gemüse.

EN: Although it was raining, we continued our journey.
DE: Obwohl es regnete, setzten wir unsere Reise fort.

EN: What is your favorite programming language?
DE: Was ist Ihre Lieblings-Programmiersprache?

EN: The meeting will start at 10 AM tomorrow.
DE: Das Treffen beginnt morgen um 10.00 Uhr.

EN: Data science involves statistics and programming.
DE: Die Datenwissenschaft umfasst Statistiken und Programmierung.

EN: He plays cricket and also studies well.
DE: Er spielt Cricket und studiert auch gut.

EN: If you work hard, you will succeed.
DE: Wenn du hart arbeitest, wirst du Erfolg haben.

EN: This smartphone has advanced features.
DE: Dieses Smartphone hat erweiterte Funktionen.

EN: Let us discuss the project later.
DE: Lassen Sie uns das Projekt später besprechen.

EN: Do you understand this concept?
DE: Verstehen Sie dieses Konzept?
```

```
print("\n--- English to Hindi ---\n")

for sent in sentences:
    translated = translate(sent, tokenizer_en_hi, model_en_hi)
    print("EN:", sent)
    print("HI:", translated)
    print()
```

```
--- English to Hindi ---

EN: I am learning artificial intelligence.
HI: मैं कृत्रिम बुद्धि सीख रहा हूँ.

EN: The weather is very pleasant today.
HI: मौसम आज बहुत सुखद है.

EN: She completed her assignment on time.
HI: उसने समय पर अपनी नियुक्ति पूरी की ।

EN: Can you help me with this problem?
HI: आप इस समस्या के साथ मेरी मदद कर सकते हैं?

EN: Machine learning is transforming industries.
HI: मशीन पढ़ना उद्योगों को परिवर्तित कर रहा है.

EN: I went to the market and bought vegetables.
HI: मैं बाजार गया और सब्ज़ियाँ खरीदा.

EN: Although it was raining, we continued our journey.
HI: हालाँकि बारिश हो रही थी, फिर भी हमने अपना सफर जारी रखा ।

EN: What is your favorite programming language?
HI: अपने पसंदीदा प्रोग्रामिंग भाषा क्या है?

EN: The meeting will start at 10 AM tomorrow.
HI: बैठक 10 कल 10 बजे शुरू होगा.

EN: Data science involves statistics and programming.
HI: डाटा विज्ञान में सांख्यिकी और प्रोग्रामिंग शामिल हैं.

EN: He plays cricket and also studies well.
HI: उन्होंने कहा कि क्रिकेट खेलता है और भी अच्छी तरह से अध्ययन.

EN: If you work hard, you will succeed.
HI: अगर आप मेहनत करते हैं, तो आप ज़रूर कामयाब होंगे ।

EN: This smartphone has advanced features.
```

```
HI: यह स्मार्टफोन की विशेषताओं को विस्तृत किया गया है.

EN: Let us discuss the project later.
HI: आइए इस बारे में बाद में चर्चा करें ।

EN: Do you understand this concept?
HI: क्या आप इस धारणा को समझते हैं?
```

```
results = []

for sent in sentences:
    fr = translate(sent, tokenizer_en_fr, model_en_fr)
    de = translate(sent, tokenizer_en_de, model_en_de)
    hi = translate(sent, tokenizer_en_hi, model_en_hi)

    results.append((sent, fr, de, hi))

# Print nicely
for r in results:
    print("EN:", r[0])
    print("FR:", r[1])
    print("DE:", r[2])
    print("HI:", r[3])
    print("-" * 50)
```

```
DE: Können Sie mir bei diesem Problem helfen?
HI: आप इस समस्या के साथ मेरी मदद कर सकते हैं?
--------------------------------------------------
EN: Machine learning is transforming industries.
FR: L'apprentissage automatique transforme les industries.
DE: Maschinelles Lernen verändert die Industrie.
HI: मशीन पढ़ना उद्योगों को परिवर्तित कर रहा है.
--------------------------------------------------
EN: I went to the market and bought vegetables.
FR: Je suis allé au marché et j'ai acheté des légumes.
DE: Ich ging auf den Markt und kaufte Gemüse.
HI: मैं बाजार गया और सब्ज़ियाँ खरीदा.
--------------------------------------------------
EN: Although it was raining, we continued our journey.
FR: Même s'il pleuvait, nous avons poursuivi notre voyage.
DE: Obwohl es regnete, setzten wir unsere Reise fort.
HI: हालाँकि बारिश हो रही थी, फिर भी हमने अपना सफर जारी रखा ।
--------------------------------------------------
EN: What is your favorite programming language?
FR: Quel est votre langage de programmation préféré?
DE: Was ist Ihre Lieblings-Programmiersprache?
HI: अपने पसंदीदा प्रोग्रामिंग भाषा क्या है?
--------------------------------------------------
EN: The meeting will start at 10 AM tomorrow.
FR: La réunion commencera demain à 10 heures.
DE: Das Treffen beginnt morgen um 10.00 Uhr.
HI: बैठक 10 कल 10 बजे शुरू होगा.
--------------------------------------------------
EN: Data science involves statistics and programming.
FR: La science des données implique des statistiques et des programmes.
DE: Die Datenwissenschaft umfasst Statistiken und Programmierung.
HI: डाटा विज्ञान में सांख्यिकी और प्रोग्रामिंग शामिल हैं.
--------------------------------------------------
EN: He plays cricket and also studies well.
FR: Il joue au cricket et étudie aussi bien.
DE: Er spielt Cricket und studiert auch gut.
```

```
EN: Do you understand this concept?
FR: Comprenez-vous ce concept ?
DE: Verstehen Sie dieses Konzept?
HI: क्या आप इस धारणा को समझते हैं?
--------------------------------------------------
```

```
complex_sentences = [
    "Although the project was difficult, we managed to complete it successfully.",
    "If you do not follow the instructions properly, the system may fail.",
    "Artificial intelligence, which is rapidly growing, is used in many industries.",
    "What will you do if the network connection is lost?",
    "The algorithm performs well when the dataset is large and clean."
]

print("\n--- Complex Sentence Translation (Hindi) ---\n")

for sent in complex_sentences:
    translated = translate(sent, tokenizer_en_hi, model_en_hi)
    print("EN:", sent)
    print("HI:", translated)
    print()
```

```
--- Complex Sentence Translation (Hindi) ---

EN: Although the project was difficult, we managed to complete it successfully.
HI: हालाँकि यह काम बहुत मुश्किल था, फिर भी हम इसे सफलतापूर्वक पूरा कर पाए ।

EN: If you do not follow the instructions properly, the system may fail.
HI: अगर आप निर्देशों को सही तरह से नहीं मानते, तो तंत्र असफल हो सकता है ।

EN: Artificial intelligence, which is rapidly growing, is used in many industries.
HI: कला - विज्ञान, जो तेज़ी से बढ़ती जा रही है, अनेक उद्योगों में प्रयोग किया जाता है ।

EN: What will you do if the network connection is lost?
HI: यदि नेटवर्क कनेक्शन टूट गया है तो आप क्या करेंगे?

EN: The algorithm performs well when the dataset is large and clean.
HI: एल्गोरिथ्म अच्छा काम करता है जब डेटासेट बड़ा है और साफ है.
```